



GROWTH & GH AXIS

CJC-1295 with DAC

CJC-1295 with DAC – 5 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	CJC-1295 with DAC
SYNONYMS / ALIASES	DAC:GRF; CJC-1295 DAC; drug-affinity-complex GHRH(1-29) analog
CAS NUMBER	863288-34-0 (parent sequence; with DAC modification) [VERIFY WITH COA]
MOLECULAR FORMULA	C165H269N47O46 [VERIFY WITH COA]
MOLECULAR WEIGHT	~3647 g/mol [VERIFY WITH COA]
AMINO ACID SEQUENCE	Modified GRF(1-29) backbone (D-Ala2, Gln8, Ala15, Leu27) with an added C-terminal Lys bearing a maleimidopropionic acid (MPA) drug-affinity-complex group for covalent albumin binding
SOURCE	Synthetic (solid-phase peptide synthesis with maleimide conjugation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	5 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-21 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. The maleimide group is reactive toward free thiols - avoid thiol-containing buffers.

STRUCTURAL NOTES

CJC-1295 with DAC is a synthetic analog of growth-hormone-releasing hormone (GHRH) built on the Modified GRF(1-29) backbone (substitutions D-Ala2, Gln8, Ala15, Leu27). It additionally carries a drug-affinity-complex (DAC) - a maleimidopropionic acid group attached via a C-terminal lysine - that forms a covalent bond with cysteine-34 of circulating serum albumin. This bioconjugation markedly extends the circulating half-life relative to the non-DAC variant; see the [[cjc-1295-no-dac]] datasheet for comparison.

SELECTED PEER-REVIEWED REFERENCES

1. Teichman S.L. et al. - *Prolonged stimulation of growth hormone and insulin-like growth factor I secretion by CJC-1295, a long-acting analog of GH-releasing hormone, in healthy adults* - Journal of Clinical Endocrinology & Metabolism, 2006 - DOI: 10.1210/jc.2005-1536
2. Jette L. et al. - *hGRF1-29-albumin bioconjugates activate the GRF receptor on the anterior pituitary in rats* - Endocrinology, 2005 - DOI: 10.1210/en.2004-1286

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.